

ARCHITECTURAL DESIGN PORTFOLIO — 2025

# Gichuhi Peter Mathia

Graduate Architect · Nairobi, Kenya

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ABOUT

# Design Statement



**Gichuhi Peter Mathia** is a graduate architect based in Nairobi, Kenya, with hands-on experience spanning construction sites, public-sector project delivery, and ISO-certified private practice.

His work moves between residential, mixed-use, institutional and master-planning scales — from a single courtyard house to a 30,000m² affordable housing scheme carried through to construction documentation. He is drawn to projects where technical rigour and spatial generosity are not a trade-off, and to briefs that ask architecture to do social work: housing that is dignified at low cost, and institutions that are genuinely used by the communities they serve.

*He is currently seeking a design role with an internationally practising studio, where his construction-site experience and design ambition can sit side by side.*

CURRICULUM VITAE

# Education, Experience & Skills

## EDUCATION

### Bachelor of Architectural Studies — Honours

Jomo Kenyatta University of Agriculture and Technology- Graduated June 2024.

### Bachelor of Architecture — Honours

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## EXPERIENCE

### Graduate Architect

Brookhill Project Limited – August 2026 to date.

### Junior Architect

Aaki Consultant Architects (ISO-certified) — Jan 2024 to Jun 2025

### Architectural Intern

State Department of Public Works — Jan to Mar 2023

### Architectural Intern

Modern Precast Kenya Limited — Jan to Mar 2022

## SKILLS

- ArchiCAD
- Lumion
- Model Making
- Engineering Survey
- Building Setting-Out
- Costing & Estimation

## SELECTED PROJECTS

— Affordable Housing Master Plan — full construction documentation

— Mixed-Use Development, Kilimani — high-rise retail & residential

— School of Architecture — institutional campus design

— Residential Bungalow — courtyard interior design

— Private Residence, Marsabit County — 5-bedroom commission

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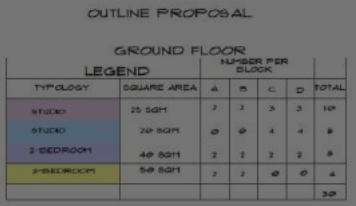
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01

# Affordable Housing Master Plan

*Nairobi County · 30,000m² · Studio to 3-bedroom duplex units across four blocks*





GICHUHI PETER MATHIA AB9211-028

# The Brief & The Plan




*Ground floor master plan — four residential blocks around a shared children's play court, with typology legend and unit counts.*

## THE PROBLEM

Deliver a high density of genuinely affordable units on a constrained urban plot, without producing the monotony that low-cost housing is usually forced into.

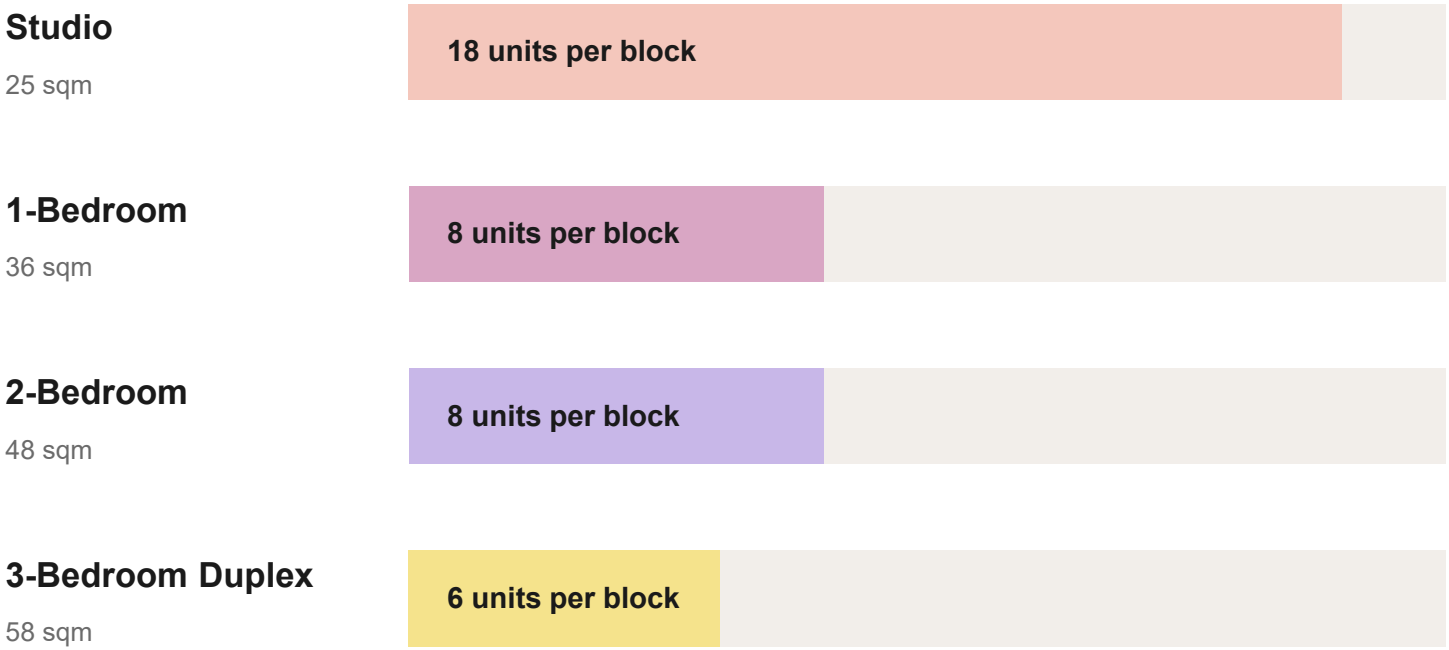
## THE STRATEGY

Four residential blocks (A–D) are arranged around a shared, car-free central court containing children's play areas, with perimeter parking and a 4m internal access road keeping vehicles out of the social heart of the scheme. A studio-to-3-bedroom unit mix, colour-coded here, lets the scheme cross-subsidise smaller units with larger ones.

## OUTCOME

Carried through from concept to full construction documentation — elevations, sections, construction details, and door/window schedules — shown on the following pages.

# Unit Typology & Distribution




*Unit mix per block (Blocks A–D) — colour-matched to the master plan legend on the previous page.*

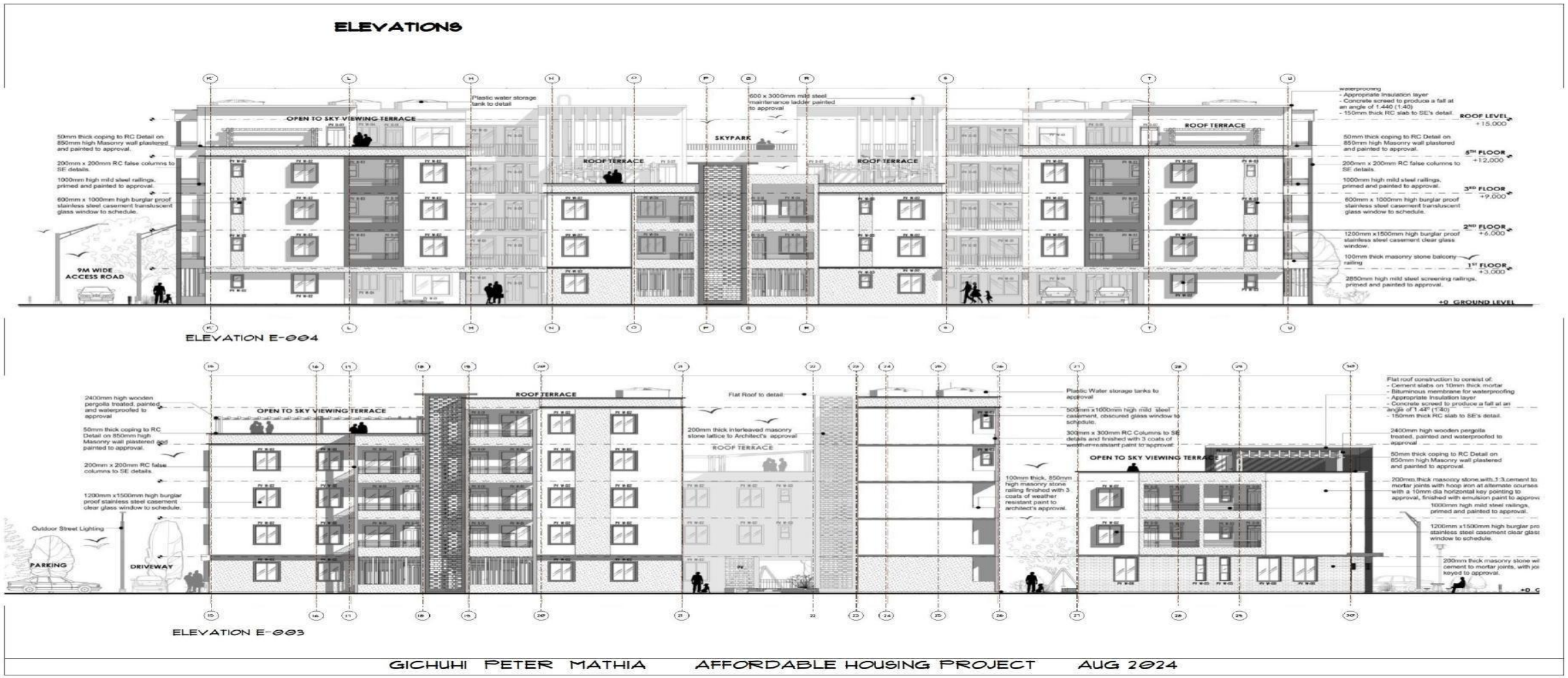
## DESIGN LOGIC

Smaller studio and one-bedroom units sit closest to the shared court and stair cores, maximising the number of households within easy reach of communal amenity. Larger duplex units are placed at block ends, giving them dual-aspect ventilation and a private ground-floor entrance separate from the shared lobby.

## CIRCULATION

Vehicles are held to the site perimeter and a single central spine; every block is reached on foot through landscaped, car-free courts — placing pedestrian safety and play space above parking convenience.

# Elevations




*East and west elevations, showing the pergola-shaded roof terraces and stone-lattice screening used to break down the block's street façade.*

# Street & Courtyard Views

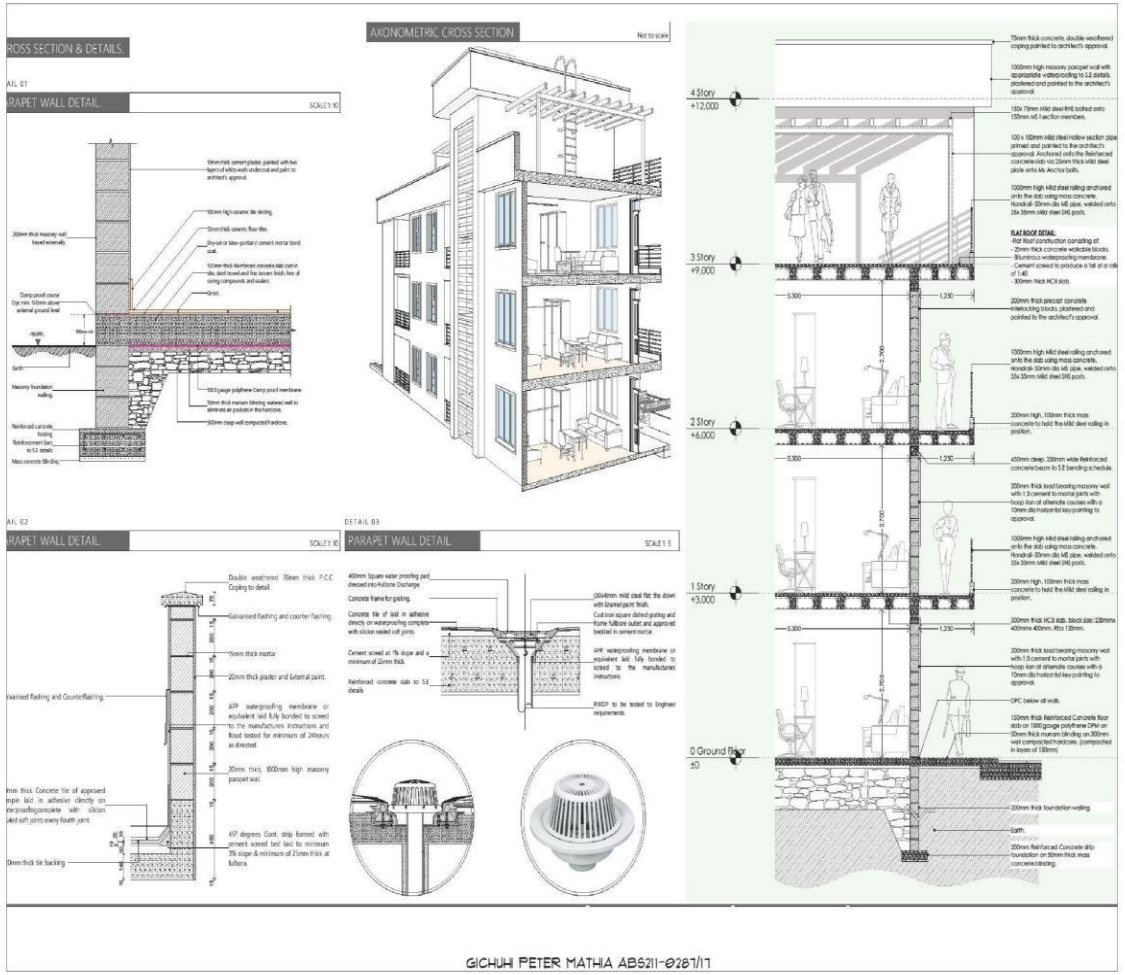



*Street-level and courtyard renders — retail frontage at grade, and a shared play court framed by the residential blocks above.*

# Physical Model & Construction Detail

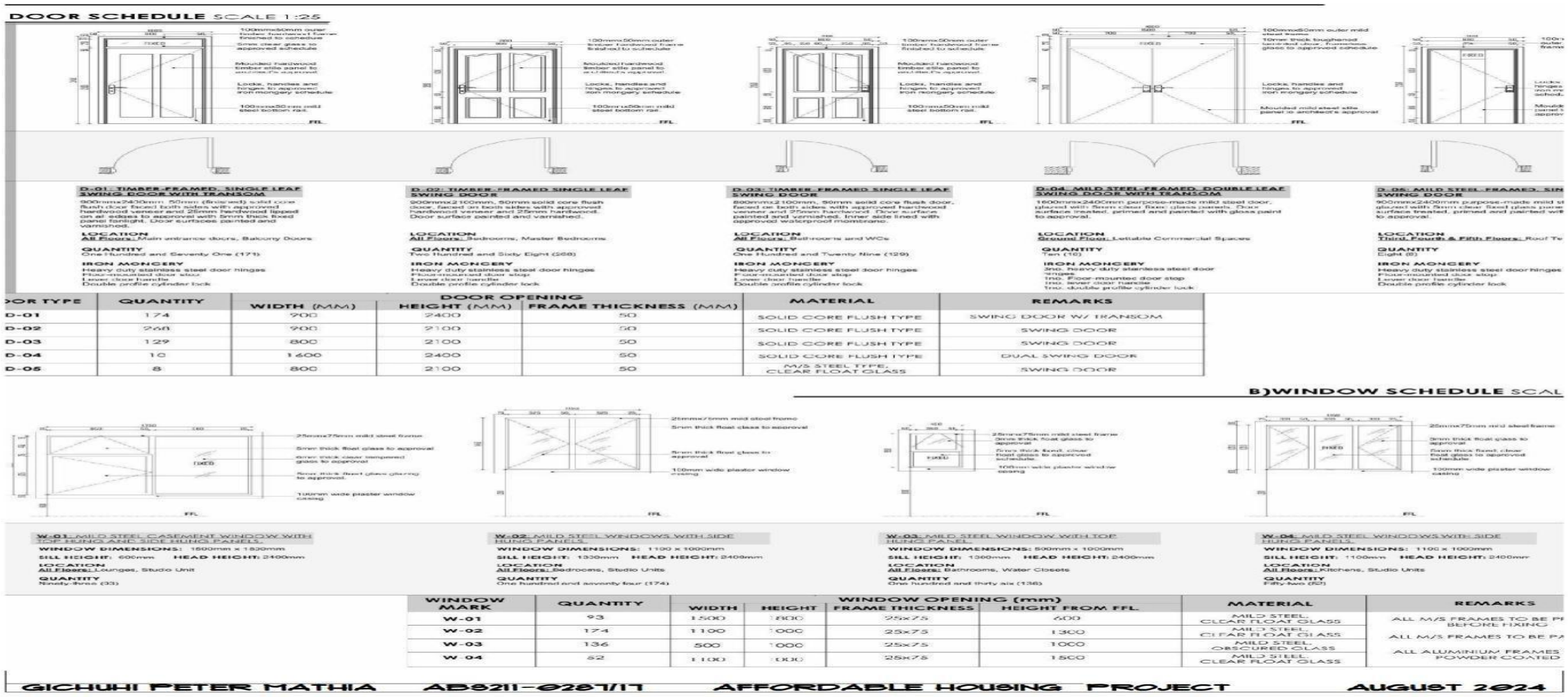


*Hand-built massing model, Blocks A–D around the shared court (parking and access road at perimeter).*




*Building section, axonometric cutaway and parapet construction details.*

# Door & Window Schedule




DOOR SCHEDULE

| DOOR TYPE | QUANTITY | DOOR OPENING WIDTH (MM) | DOOR OPENING HEIGHT (MM) | DOOR OPENING FRAME THICKNESS (MM) | MATERIAL | REMARKS |
|---|---|---|---|---|---|---|
| D-01 | 174 | 900 | 2400 | 50 | SOLID CORE FLUSH TYPE | SWING DOOR W/ TRANSOM |
| D-02 | 268 | 900 | 2100 | 50 | SOLID CORE FLUSH TYPE | SWING DOOR |
| D-03 | 129 | 800 | 2100 | 50 | SOLID CORE FLUSH TYPE | SWING DOOR |
| D-04 | 10 | 1600 | 2400 | 50 | SOLID CORE FLUSH TYPE | DUAL SWING DOOR |
| D-05 | 8 | 800 | 2100 | 50 | M/S STEEL TYPE, CLEAR FLOAT GLASS | SWING DOOR |

WINDOW SCHEDULE

| WINDOW MARK | QUANTITY | WINDOW OPENING (mm) WIDTH | HEIGHT | FRAME THICKNESS | HEIGHT FROM FFL | MATERIAL | REMARKS |
|---|---|---|---|---|---|---|---|
| W-01 | 93 | 1500 | 1800 | 25x75 | 600 | MILD STEEL CLEAR FLOAT GLASS | ALL M/S FRAMES TO BE PI BEFORE FIXING |
| W-02 | 174 | 1100 | 1000 | 25x75 | 1300 | MILD STEEL CLEAR FLOAT GLASS | ALL M/S FRAMES TO BE PA |
| W-03 | 136 | 500 | 1000 | 25x75 | 1000 | MILD STEEL OBSCURED GLASS | |
| W-04 | 52 | 1100 | 1000 | 25x75 | 1500 | MILD STEEL CLEAR FLOAT GLASS | ALL ALUMINIUM FRAMES POWDER COATED |

*Full door and window schedule — five door types and four window types, with quantities, dimensions and ironmongery specified per unit.*

02

# Mixed-Use Development

*Kilimani, Nairobi · Retail, medical, office and residential in a single tower*



# Programme & Massing



*Tower massing, viewed from street level. A double-height retail podium anchors the base of a 20-storey residential and office tower.*

## THE BRIEF

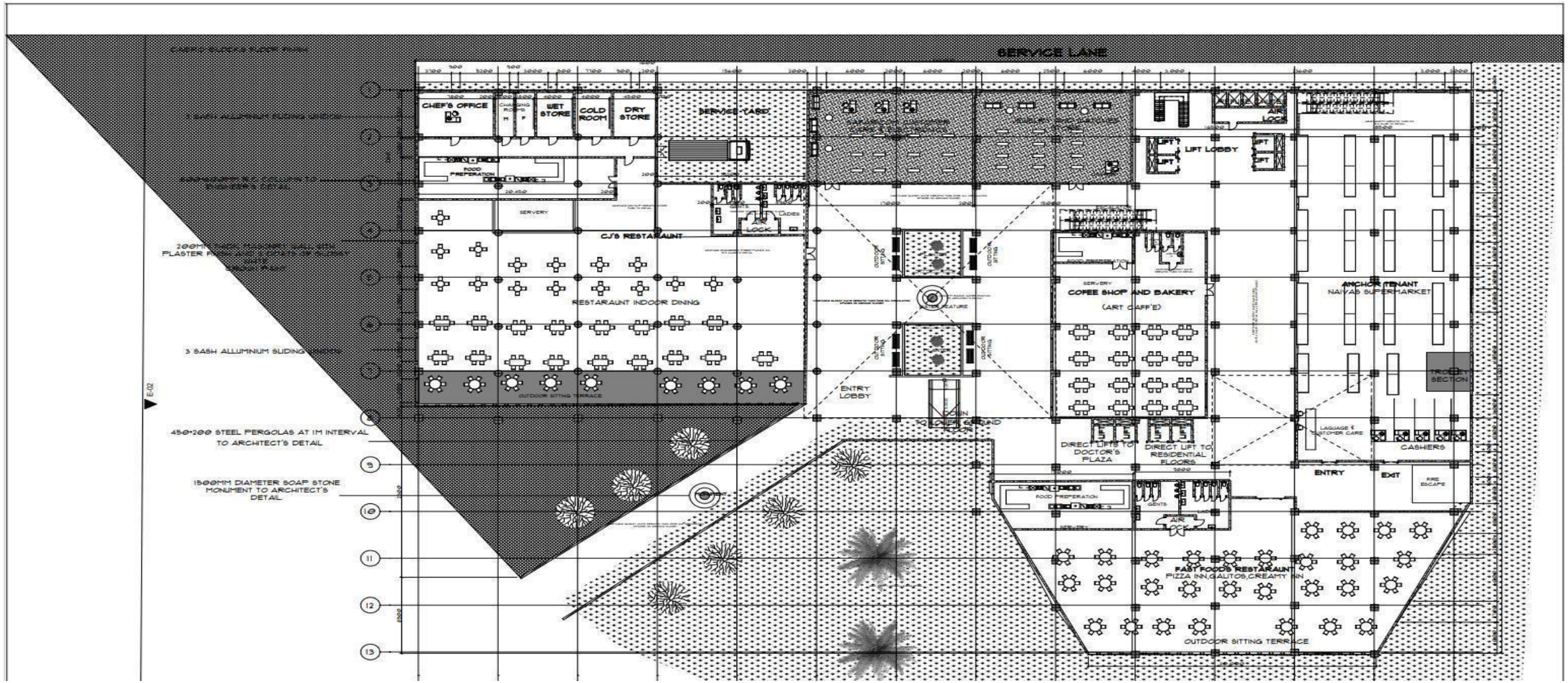
A blended commercial and residential programme for a posh Kilimani plot, following garden-city planning principles for the ground-level public realm.

## VERTICAL STACKING

- Penthouses
- 3-Bed Duplex Apartments
- 2-Bed Duplex Apartments
- 1-Bed / Studio Apartments
- Office Spaces
- Doctor's Plaza
- Restaurants & Furniture Showrooms
- Supermarket
- Basement Parking

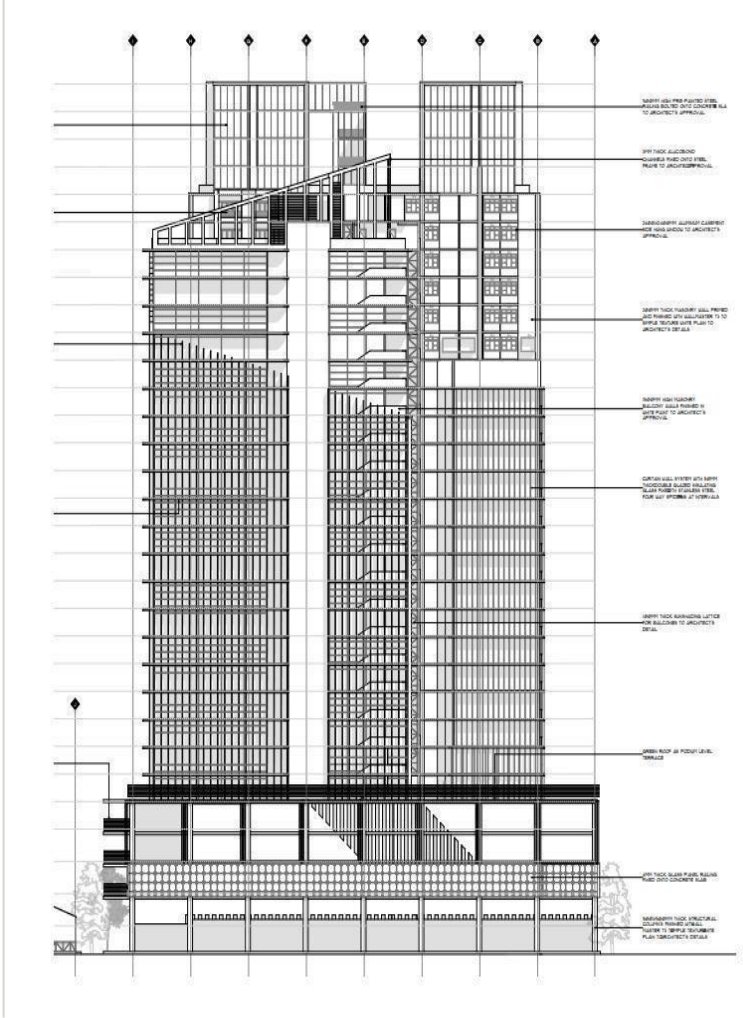
*Programme stacking, ground to roof — basement parking and retail anchor the base; residential and office uses layer above, with penthouses topping the tower.*

# Ground Floor Plan

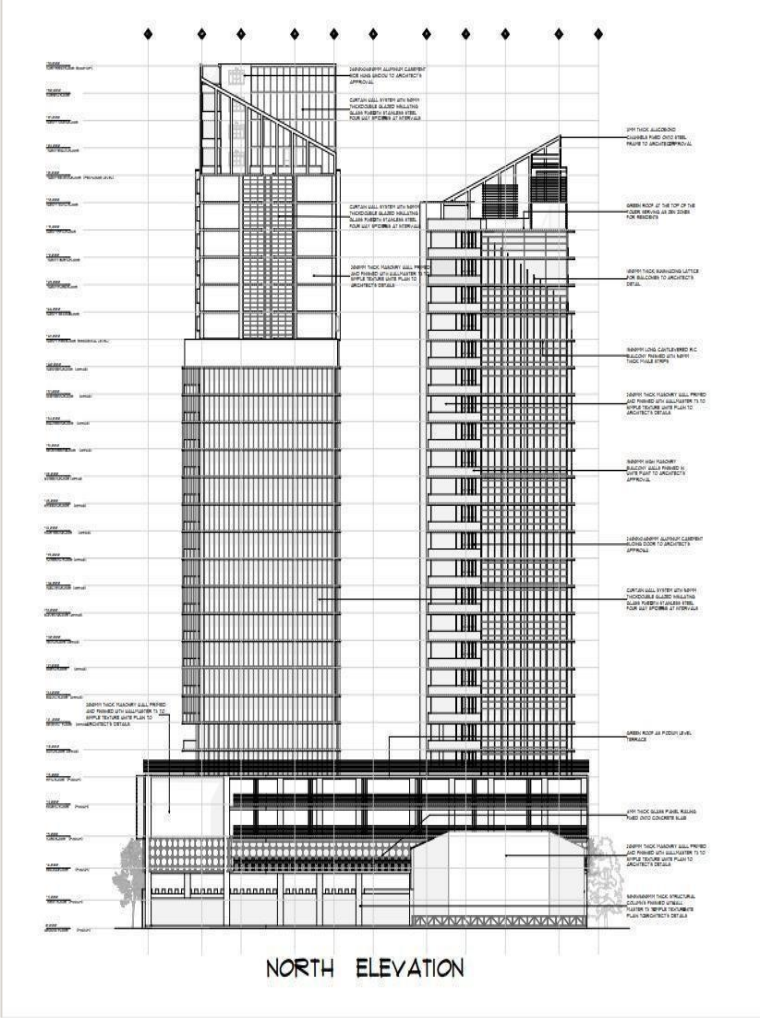



*Ground floor: restaurant and café tenancies wrap a landscaped entry lobby, with a supermarket anchor, direct lift access to the doctor's plaza and residential floors above.*

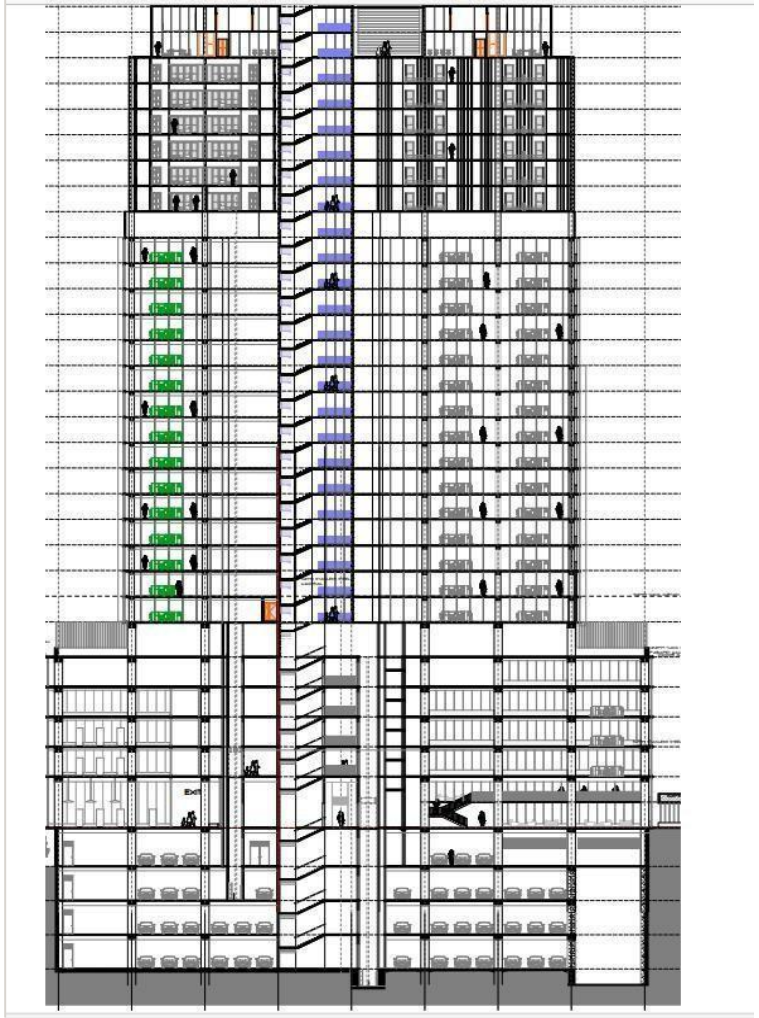
# Elevations & Section



*East elevation*




*North elevation*



*Building section*

# Renders



*Tower, viewed from the podium roof terrace.*

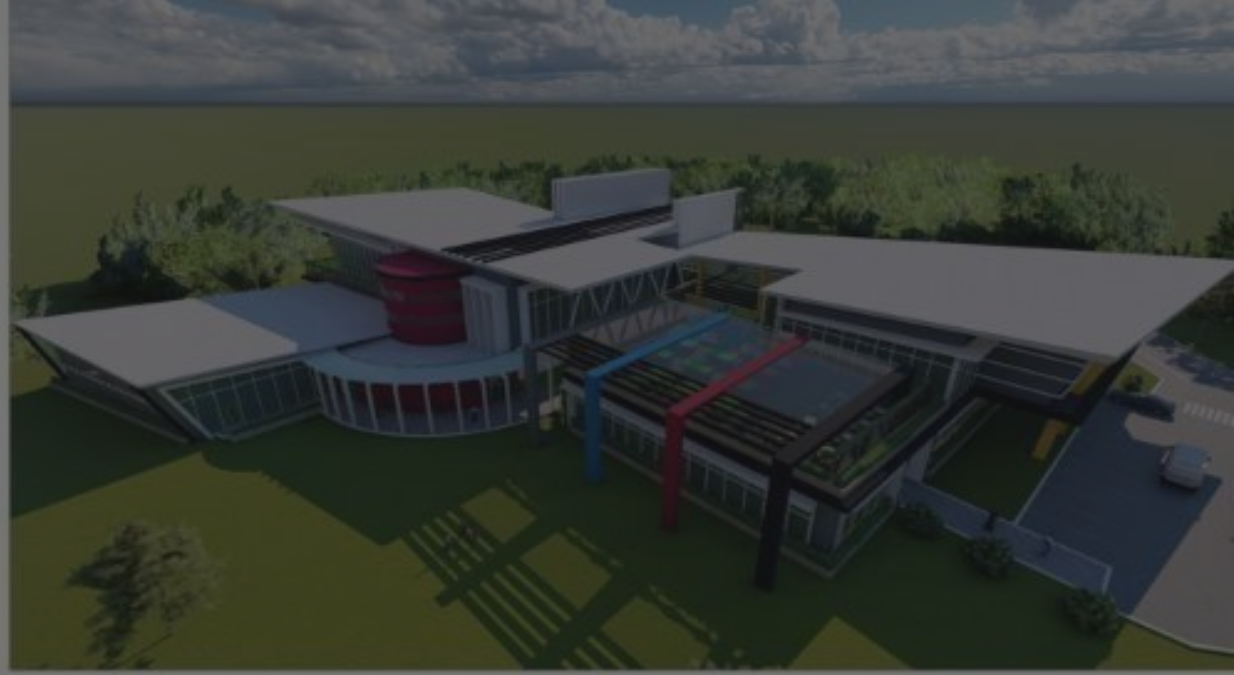


*Tower, street-level approach.*




03

WELL DEFINED ENTRANCE



STUDENT OUTDOOR RECREATION SPACE

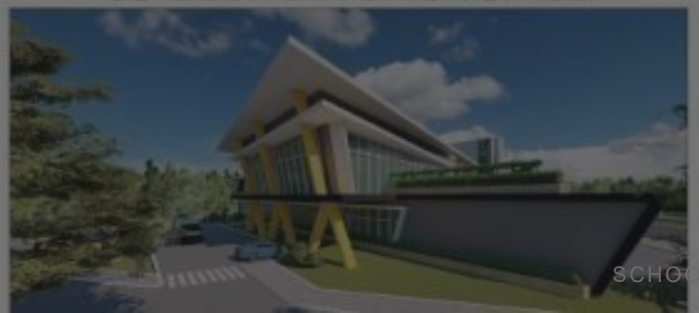
# School of Architecture

*Institutional campus · Studios, exhibition space and student outdoor recreation*





OUTDOOR EXHIBITION SPACE





# Concept & Renders




WELL DEFINED ENTRANCE



STUDENT OUTDOOR RECREATION SPACE





OUTDOOR EXHIBITION SPACE





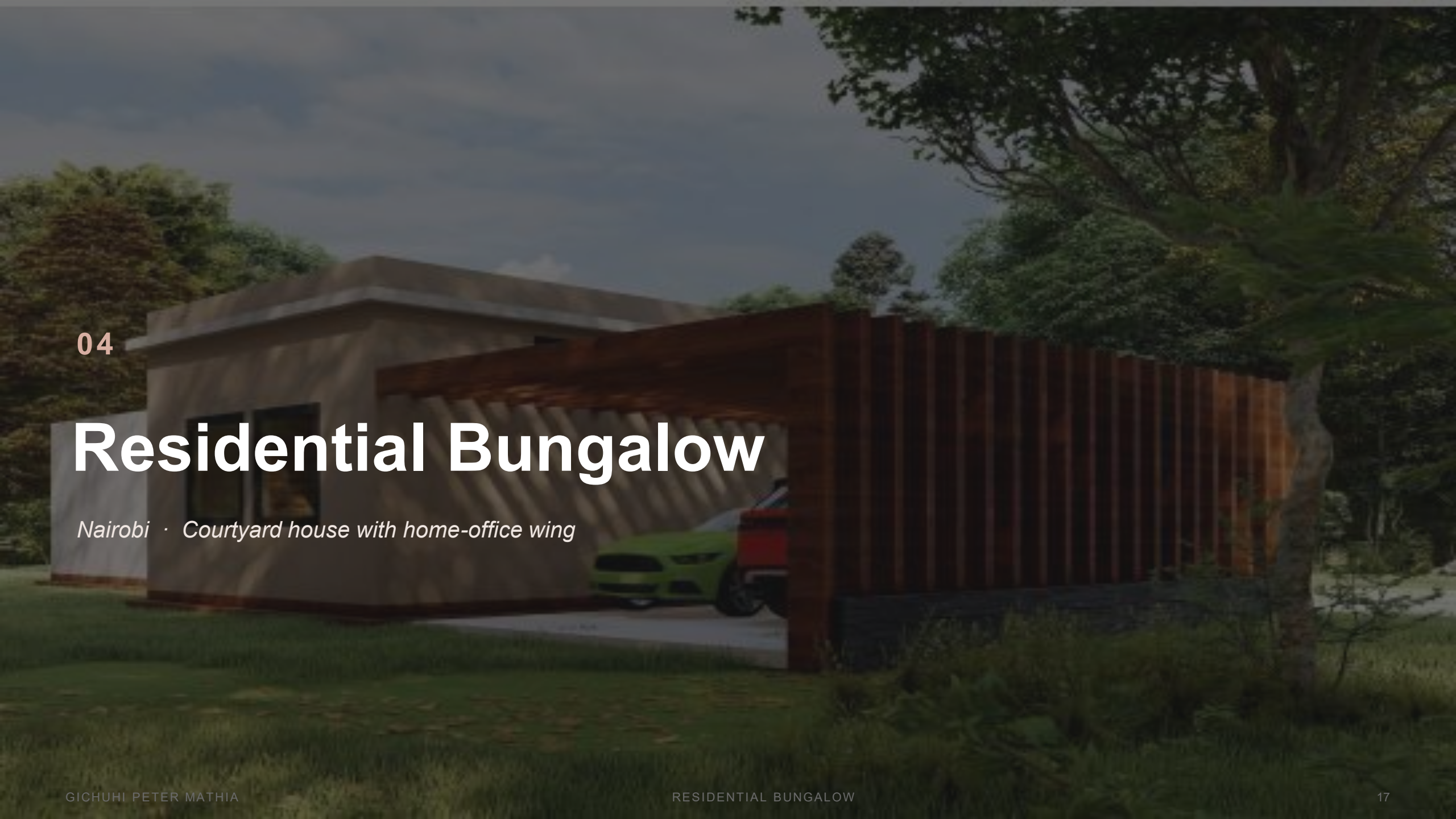
*A well-defined public entrance, an outdoor exhibition pavilion, and a covered student recreation court give the school three distinct social spaces beyond the studio floors.*

## THE IDEA

An architecture school should exhibit its own students' work as readily as it teaches — so the building is organised around a sequence of outdoor and semi-outdoor spaces for critique, exhibition and informal review, rather than corridors alone.

## MATERIALS & IDENTITY

A restrained white volume is punctuated by a signal-red drum and a yellow structural frame at the entrance — a deliberate, legible identity for a building that is, in effect, its own advertisement.



04

# Residential Bungalow

*Nairobi · Courtyard house with home-office wing*

# Concept & Plan




*Floor plan — bedroom and living wing to the north, a two-office home-work wing to the south, joined by a shaded central courtyard.*

## THE IDEA

A house carries a collective memory of comfort — warmth, calm, and a felt connection to its site. Every principal room in this plan opens onto a planted central courtyard, so daily life is organised around a piece of the outdoors rather than a corridor.

## ORIENTATION & COMFORT

Nairobi sits almost on the equator, where the sun tracks close to directly overhead year-round. The courtyard is deliberately narrow and deep rather than open and wide, so its own walls self-shade the surrounding rooms through the middle of the day, while the lounge and dining spaces on the east side catch low, gentle morning light.

# Elevations



*North elevation — split roof planes and a covered entry porch beneath mature palm and acacia planting.*



*South elevation — timber louvre screening to the home-office wing, moderating glare without closing off the courtyard view.*

# Interior



*Kitchen and dining, opening directly onto the courtyard through a full-height sliding timber screen.*



*Living room — a warm palette of timber flooring, stone cladding and soft furnishing carries the courtyard's material language indoors.*



05

# Private Residence, Marsabit County

*5-bedroom family residence · Hot, arid northern Kenya climate*

# Site & Climate Response



*Site plan — main house, staff quarters and covered parking arranged around a walled, planted forecourt.*

## CLIMATE STRATEGY

Marsabit County is hot and arid, with intense daytime solar gain and wide day-to-night temperature swings. The design responds with strategies proven across northern Kenya's vernacular and contemporary architecture alike:

- Deep, tiled roof overhangs shading walls and openings through the middle of the day
- Stone-clad ground floor for thermal mass, moderating internal temperature swings
- Arcaded verandas on two elevations, buffering habitable rooms from direct sun
- A high perimeter wall and mature planting reducing dust and wind loading at ground level

# Exterior



*Aerial approach view — the main residence, guest wing and staff quarters share a single tiled-roof language around a landscaped forecourt.*

# Interior — Entrance Lobby



*Entrance lobby.*

## THE SPACE

A pitched timber ceiling rises over a lobby opened to the garden through a run of arches, carrying the exterior's arcaded language indoors.

## MATERIALS

Dark timber ceiling, white plastered walls and a stone-clad planter wall set into the arches.

# Interior — Family Living Space



*Family living space.*

## THE SPACE

A double-height room with an upper-level landing overlooking it, organised around a central seating group on a round rug, with large steel-framed windows bringing in light.

## MATERIALS

Timber slatted screens, a stone-clad feature wall, timber flooring and dark steel-framed glazing.

# Interior — Living Room, Ground Floor



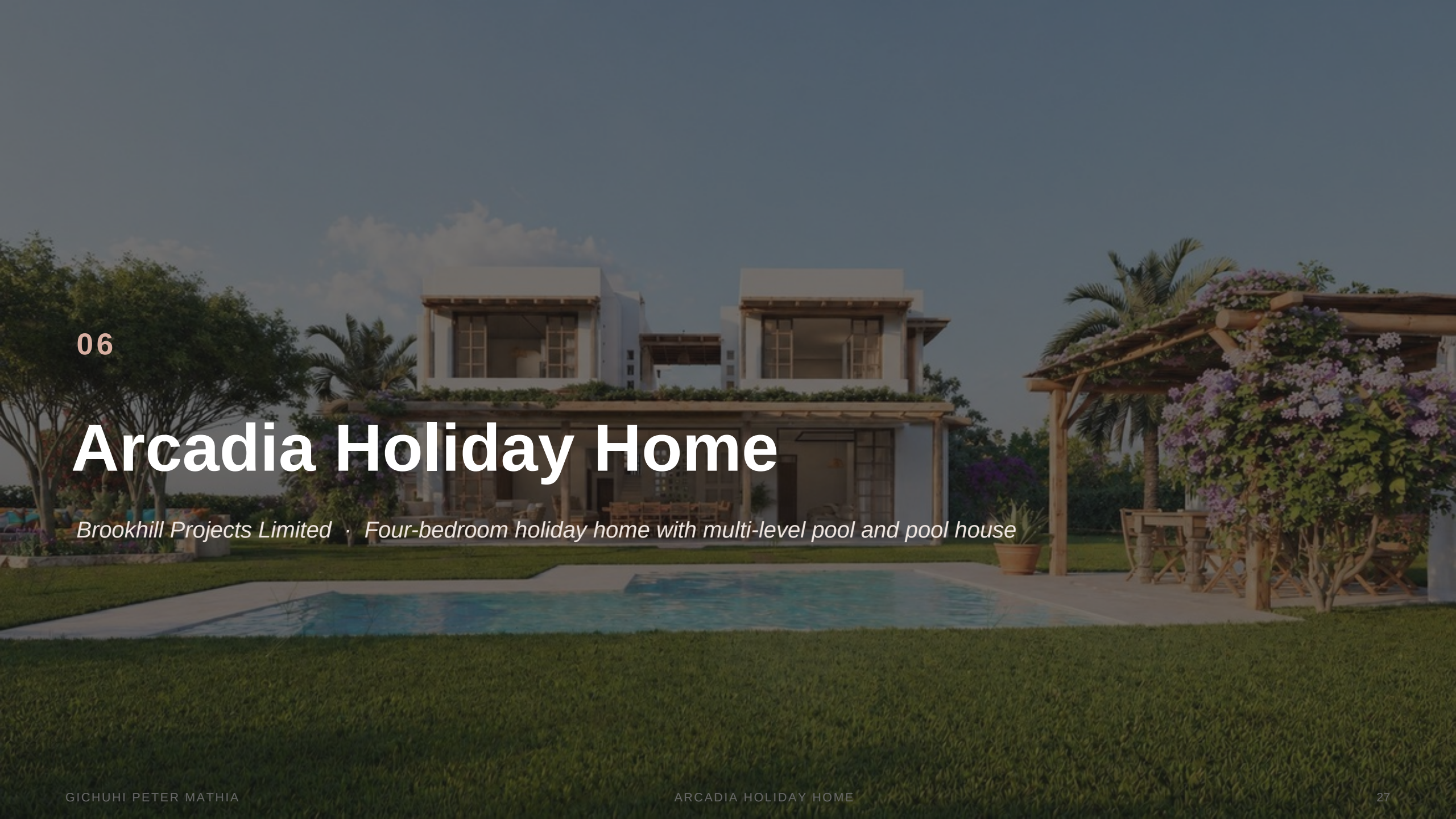
*Living room, ground floor.*

## THE SPACE

A tall ground-floor living room built around a fireplace on a stone-clad chimney breast, with seating grouped around a central table and tall timber doors to either side.

## MATERIALS

Timber-slat ceiling, timber flooring, a stone-clad chimney and dark timber doors.



06

# Arcadia Holiday Home

*Brookhill Projects Limited · Four-bedroom holiday home with multi-level pool and pool house*

# The Brief & The Site




*Site plan — gate and staff area at the entrance, the house mid-plot, the pool and cabana beyond.*

## LEGEND

1. Gate
2. Driveway & Parking
3. Store & Staff WC
4. Staff break-out space
5. Staff resting bench
6. House
7. Under-tree seating
8. Pool
9. Pool cabana

## THE BRIEF

A family holiday home on a hedged plot: four bedrooms, a quiet AC-cooled office, an enclosed kitchen beside the social spaces, a sealed family room and a large covered terrace kept as open as possible.

## THE SITE

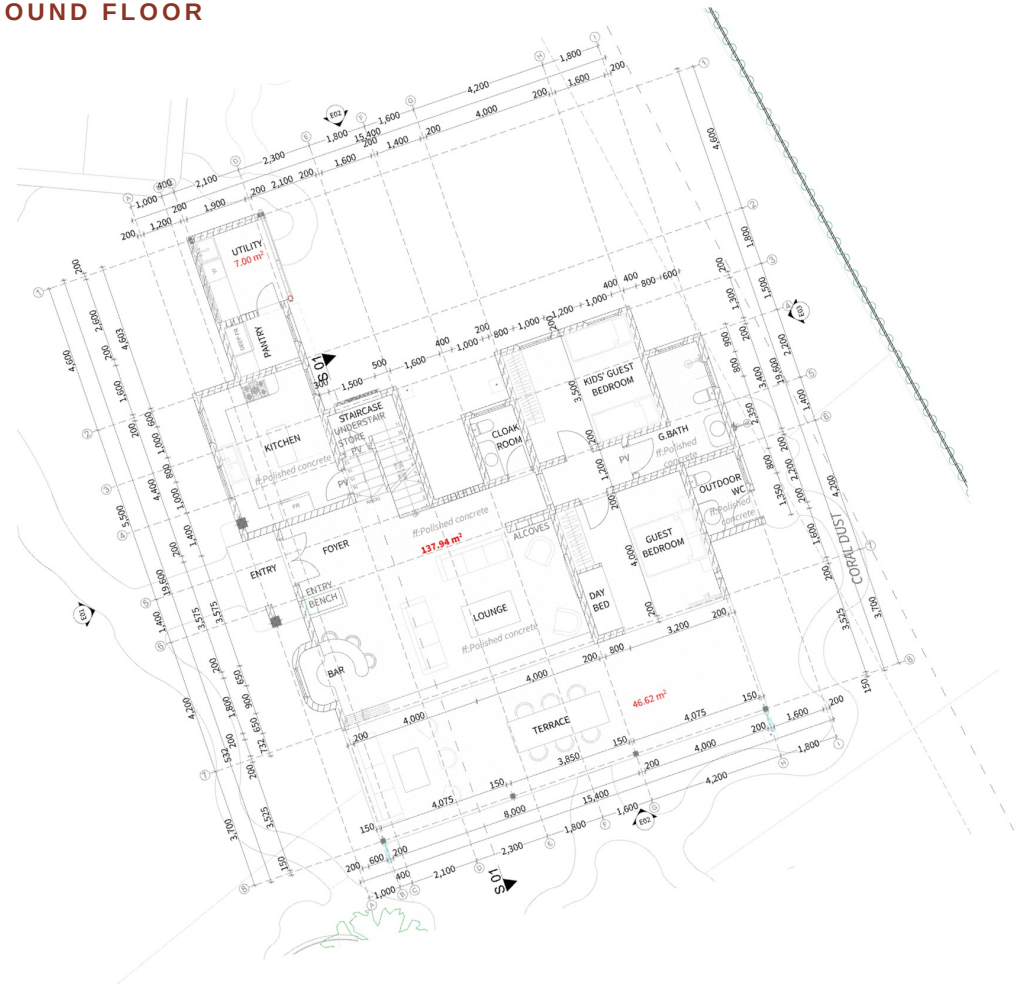
Gate, driveway and parking, a store with staff WC, a break-out space and a resting bench are grouped at the entrance. The house sits mid-plot, with under-tree seating, a multi-level pool and a pool cabana beyond it.

## THE HOUSE

Two white-plastered volumes under clay-tile pitches, linked at the centre. The terrace, pool and garden open off the ground floor; roof terraces take social use and services.

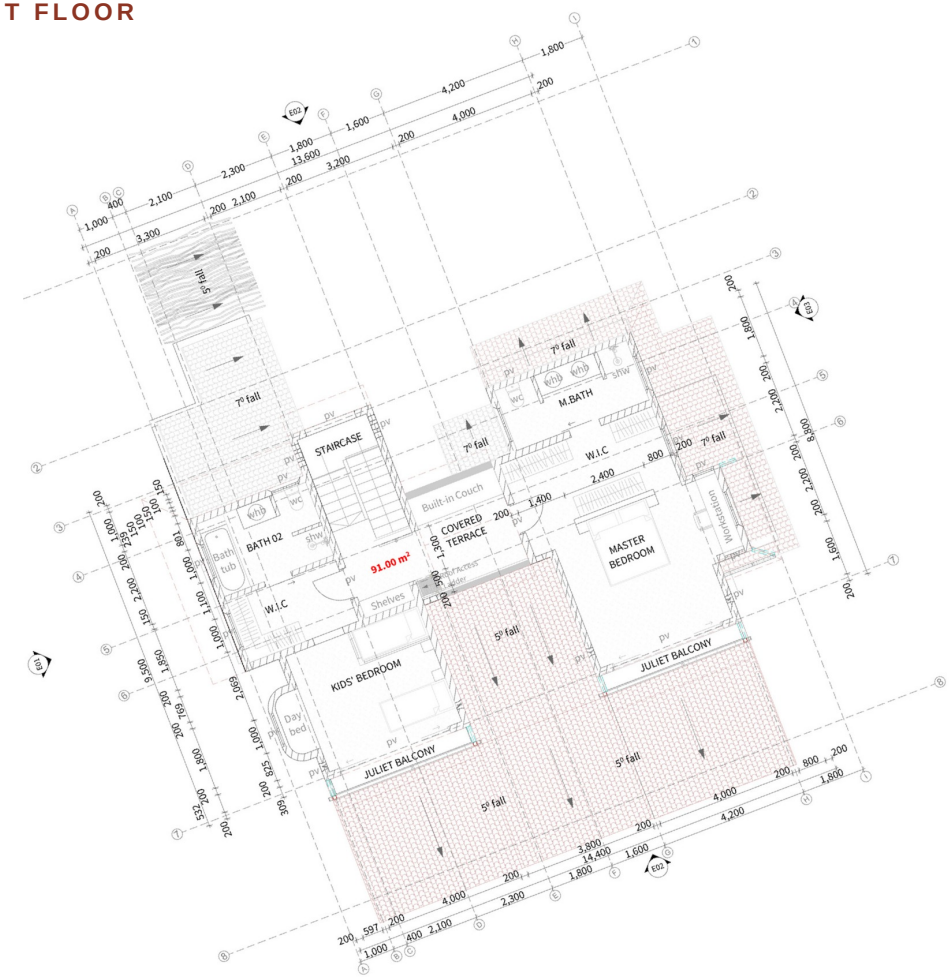
# Ground & First Floor Plans

## GROUND FLOOR




*Ground floor — foyer, lounge and bar open onto the covered terrace; kitchen, utility and pantry are enclosed rather than open-plan; two guest bedrooms, a bath and an outdoor WC.*

## FIRST FLOOR




*First floor — master suite with bath, walk-in closet and workstation, and a children's bedroom with its own bath and walk-in closet, linked by a covered terrace.*

# Approach & Entrance



### THE APPROACH

A gravel driveway and parking court, screened by clipped hedges, make a quiet arrival; the white house is glimpsed through planting rather than presented head-on.

*Driveway view.*



### THE ENTRANCE

A rendered white wall is pierced by a feature arched timber door and timber-shuttered windows, with projecting timber members shading the upper floor.

*Front entrance view — arched timber door, shuttered windows, timber sun-shading.*

# Pool & Terrace



*Pool-side view.*

## POOL SIDE

The ground floor opens to the pool and garden behind a continuous timber pergola, planted along its edge, with the first-floor bedrooms set back above.



*Terrace view — indoor-outdoor living, shaded by natural-textile canopies.*

## COVERED TERRACE

A large, airy terrace — weather-protected but as open as possible — carries lounge and dining space that flows into the house, with textile canopies for adjustable shade.

# Garden Pavilions



## POOL CABANA

A poolside dining pavilion on a timber frame, roofed in sisal poles and hung with sheer curtains, with a bougainvillea trellis growing over it.

*Pool cabana — sisal-pole roof and bougainvillea trellis.*



## UNDER-TREE SEATING

A curved masonry bench wraps a pair of multi-stem trees, with African-print cushions, a planted bed at the roots and festoon lighting for the evening.

*Under-tree seating.*

# Roof Terraces & Staff Areas




## ROOF TERRACES

Two flat roofs sit between the clay-tile pitches: a social terrace, and a services terrace that screens the solar water heaters, loft tank and covered AC units.

*Roof terraces view.*



## STAFF UTILITY AREA

A pergola-covered outdoor laundry with a built-in sink and shelves, and a picnic bench for breaks, set near the gate and apart from the house.

*Staff utility area.*

# Interior — Ground Floor



*Lounge*



*Bar*



*Kitchen*



*Guest bedroom*

## GROUND FLOOR

Lounge, bar and dining share one beamed volume that runs out to the covered terrace; the kitchen stays enclosed, with built-in plastered joinery and shuttered windows.

## MATERIALS

Exposed timber ceiling beams, plastered walls, concrete floors, rattan and timber furniture, and African-print textiles.

# Interior — Children's Suite



*Children's bedroom*



*Balcony doors and window seat*



*Walk-in closet*



*Bath*

## CHILDREN'S SUITE

Bunk and single beds, a window seat in African-print fabric and built-in plastered shelving, opening through timber-framed glazed doors to a balcony.

## WALK-IN CLOSET & BATH

A plastered walk-in closet with timber shelves and hanging rails, and a bath with tub, basin and a curved screen wall around the shower.

# Interior — Master Suite



*Master bedroom*



*Master bedroom — workstation*



*Walk-in closet*



*Master bath*

## MASTER SUITE

A sheer-curtained sleeping alcove opens through timber-framed glazed doors to a balcony; a desk corner serves as the temporary home office the brief asked for.

## WALK-IN CLOSET & BATH

Plastered shelving and hanging rails in the walk-in closet, and a twin-basin vanity with a plastered surround in the master bath.

SKILLS & SOFTWARE

# Tools of the Trade

## DESIGN & DRAFTING

- ArchiCAD
- Model Making

## VISUALISATION

- Lumion

## SITE & TECHNICAL

- Engineering Survey
- Building Setting-Out
- Costing & Estimation

**A note on BIM & software depth**

*This portfolio currently demonstrates ArchiCAD through finished drawing output rather than model views. Applicants targeting BIM-specific roles should add clash-detection, schedule-automation, or model-navigation screenshots directly from ArchiCAD to substantiate this skill visually.*

THANK YOU

# Referees & Contact

**Architect George Njeru**

Registered & Practising Architect, Kenya · contact details available on request

**Architect Josephine**

Architect & Lecturer, JKUAT · contact details available on request

**Architect Kelvin Kariuki**

Architect, Aaki Consultants · contact details available on request

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